

Radian Measure

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Up to now, all angle measures in this text have been given in degrees. This well-known unit for expressing angles is often found in applications such as navigation and surveying. However, in mathematics and other related fields, an alternative form of angle measure, radian measure, is more frequently used.

Radian measure is based on a circular perspective, where the amount of rotation defined by a central angle (angle with vertex at the center of the circle) is a direct comparison between the radius of the circle and the length along the intercepted arc (the portion of the circle between the initial and terminal side of an angle). For example, in a circle with a radius of 1 unit, a central angle measuring 1 radian will intercept an arc of length 1 unit. Likewise, a central angle measuring 2 radians will intercept an arc with a length of 2 units, etc. **As a rule, any angle measure where units are not explicitly stated is generally assumed to be in radians.**

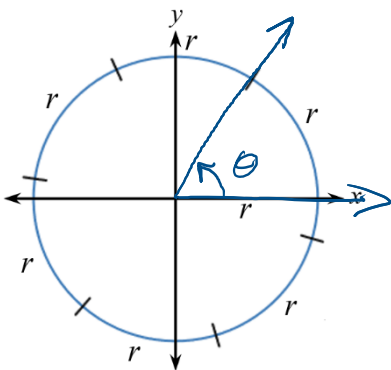
DEFINITION: *The Radian*

A radian is an angle, which, if placed with vertex at the center of a circle, will intercept an arc with a length equal to the radius of that circle.

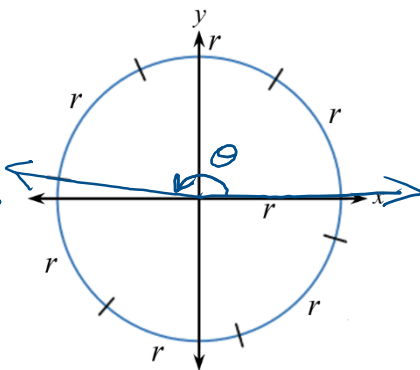
EXAMPLE 1

Graph each angle.

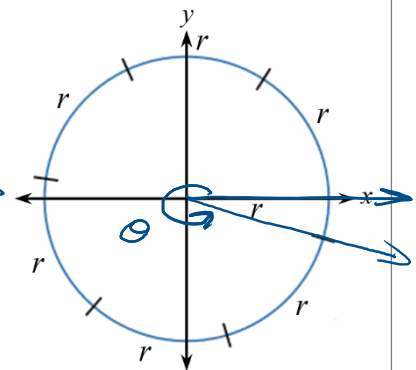
(a) $\theta = 1$



(b) $\theta = 3$



(c) $\theta = 6$



Radian-Degree Conversion

Looking back at Example 1, we can see that the angle $\theta = 3$ represents a measure that is nearly 180° , and $\theta = 6$ is just shy of one complete rotation or 360° . These observations are confirmed by the angle relationships below.

To convert between degree and radian measure, we often reference two well-known angles:

$$\underline{360} \text{ degrees} = \underline{2\pi} \text{ radians} \approx 6.28 \text{ radians}$$

$$\underline{180} \text{ degrees} = \underline{\pi} \text{ radians} \approx 3.14 \text{ radians}$$

Using either of these equations, we can create a “unit ratio” that will allow us to easily convert from one system of angle measure to the other.

$$\frac{180^\circ}{\pi} = \frac{\pi}{180^\circ} = 1$$

PROCEDURE: Radian-Degree Conversion

To convert from degree to radian measure, multiply by $\frac{\pi}{180^\circ}$.

Note the above unit ratio will multiply “in” the radian measure and divide “out” the degree measure.

To convert from radian to degree measure, multiply by $\frac{180^\circ}{\pi}$.

Note the above unit ratio will multiply “in” the degree measure and divide “out” the radian measure.

EXAMPLE 2

Convert each angle measure from degrees to radians or from radians to degrees.

$$\begin{aligned} \text{(a) } 225^\circ & \left(\frac{\pi}{180^\circ} \right) \\ &= \frac{225^\circ}{180^\circ} \pi = \frac{5}{4} \pi = \frac{5\pi}{4} \end{aligned}$$

$$\begin{aligned} \text{(b) } -60^\circ & \left(\frac{\pi}{180^\circ} \right) \\ &= \frac{-60^\circ}{180^\circ} \pi = -\frac{1}{3} \pi = -\frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{(c) } \frac{3\pi}{4} & \left(\frac{180^\circ}{\pi} \right) \\ &= \frac{3}{4} (180^\circ) = 135^\circ \end{aligned}$$

$$\begin{aligned} \text{(d) } \frac{7\pi}{12} & \left(\frac{180^\circ}{\pi} \right) \\ &= \frac{7}{12} (180^\circ) = 105^\circ \end{aligned}$$

Reference Angles and Radian Measure

PROCEDURE: Finding the Reference Angle for any Non-Quadrantal Angle

Given a non-quadrantal angle θ , in standard form, θ_r can be found using the following procedure.

Step 1: If $\theta > 2\pi$ or if $\theta < 0$, find a coterminal angle by adding or subtracting 2π as many times as needed to get an angle greater than 0 but less than 2π .

Step 2: If the result from Step 1 is between 0 and $\pi/2$, then this result is your reference angle. If not, subtract π and continue to Step 3.

Step 3: If the result from Step 2 is between 0 and $\pi/2$, then this result is your reference angle. If this result is between $-\pi/2$ and 0, then the absolute value of this result is your reference angle. If not, subtract π and repeat this step.

EXAMPLE 3

For each angle given here, determine the reference angle, if possible.

$$\begin{aligned}\text{(a)} \theta &= \frac{5\pi}{3} = \frac{5}{3}\pi - \frac{3}{3}\pi \\ &= \frac{2}{3}\pi - \frac{3}{3}\pi \\ &= -\frac{1}{3}\pi \rightarrow \theta_r = \frac{\pi}{3}\end{aligned}$$

$$\begin{aligned}\text{(b)} \theta &= \frac{7\pi}{6} = \frac{7}{6}\pi - \frac{6}{6}\pi \\ &= \frac{1}{6}\pi \rightarrow \theta_r = \frac{\pi}{6}\end{aligned}$$

$$\begin{aligned}\text{(c)} \theta &= \frac{3\pi}{4} = \frac{3}{4}\pi - \frac{4}{4}\pi \\ &= -\frac{1}{4}\pi \rightarrow \theta_r = \frac{\pi}{4}\end{aligned}$$

$$\begin{aligned}\text{(d)} \theta &= 3\pi \leftarrow \text{quadrantal angle!} \\ \theta_r & \text{ DNE}\end{aligned}$$

$$\begin{aligned}\text{(c)} \theta &= \frac{8\pi}{3} = \frac{8}{3}\pi - \frac{6}{3}\pi \\ &= \frac{2}{3}\pi - \frac{3}{3}\pi \\ &= -\frac{1}{3}\pi \rightarrow \theta_r = \frac{\pi}{3}\end{aligned}$$

$$\begin{aligned}\text{(d)} \theta &= -\frac{19\pi}{6} = -\frac{19}{6}\pi + \frac{12}{6}\pi \\ &= -\frac{7}{6}\pi + \frac{12}{6}\pi \\ &= \frac{5}{6}\pi - \frac{6}{6}\pi \\ &= -\frac{1}{6}\pi \rightarrow \theta_r = \frac{\pi}{6}\end{aligned}$$

EXAMPLE 4

Using the space provided, construct a first quadrant chart using radian measure.

	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin \theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undef.

EXAMPLE 5

Continuing from Example 3, evaluate each expression below using reference angles.

$$(a) \sin\left(\frac{5\pi}{3}\right) \quad \frac{3}{2}\pi < \frac{5}{3}\pi < 2\pi$$

$$\frac{5\pi}{3} \in QIV, \sin\left(\frac{5\pi}{3}\right) < 0$$

$$\theta = \frac{5\pi}{3} \rightarrow \theta_r = \frac{\pi}{3}$$

$$\sin\left(\frac{5\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$(b) \cos\left(\frac{7\pi}{6}\right) \quad \pi < \frac{7}{6}\pi < \frac{3}{2}\pi$$

$$\frac{7\pi}{6} \in QIII, \cos\left(\frac{7\pi}{6}\right) < 0$$

$$\theta = \frac{7\pi}{6} \rightarrow \theta_r = \frac{\pi}{6}$$

$$\cos\left(\frac{7\pi}{6}\right) = -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$$

$$(c) \tan\left(\frac{3\pi}{4}\right) \quad \frac{1}{2}\pi < \frac{3}{4}\pi < \pi$$

$$\frac{3\pi}{4} \in QII, \tan\left(\frac{3\pi}{4}\right) < 0$$

$$\theta = \frac{3\pi}{4} \rightarrow \theta_r = \frac{\pi}{4}$$

$$\tan\left(\frac{3\pi}{4}\right) = -\tan\left(\frac{\pi}{4}\right) = -1$$

$$(d) \csc\left(\frac{8\pi}{3}\right) \quad \frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3} \text{ (coterminal)}$$

$$\frac{\pi}{2} < \frac{2}{3}\pi < \pi \text{ (QII)}$$

$$\frac{8\pi}{3} \in QII, \csc\left(\frac{8\pi}{3}\right) > 0$$

$$\theta = \frac{8\pi}{3} \rightarrow \theta_r = \frac{\pi}{3}$$

$$\csc\left(\frac{8\pi}{3}\right) = \csc\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{\sin\left(\frac{\pi}{3}\right)} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

